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MAGNETIC DISK DEVICE

Abstract

According to one embodiment, a magnetic disk device includes a magnetic disk, a gimbal, and a controller. The magnetic disk rotates around a first axis. The gimbal includes a magnetic head and a piezoelectric element. The piezoelectric element is deformed in accordance with an applied drive voltage to move the magnetic head. The controller applies a polarization voltage to the piezoelectric element to give spontaneous polarization to the piezoelectric element. The polarization voltage satisfies a relationship of $\Delta V = (2\pi f)A \cdot \Delta T < \Delta V_{\max}$, where “ ΔV ” is a voltage change amount of the polarization voltage per unit time, “ f ” is a frequency of the polarization voltage, “ A ” is an amplitude of the polarization voltage, “ ΔT ” is a time resolution of the polarization voltage, and “ $\Delta V_{\max}$ ” is a voltage change amount per unit time that excites resonance of the gimbal in application to the piezoelectric element.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-029878, filed on Feb. 29, 2024; the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a magnetic disk device.

BACKGROUND

[0003] A magnetic disk device, such as a hard disk drive, is provided with a gimbal, for example. The gimbal includes a part of a suspension, and a magnetic head and a piezoelectric element that are mounted on the suspension. When the piezoelectric element is supplied with a voltage, this element is deformed due to the inverse piezoelectric effect and adjusts the position of the magnetic head.

[0004] The spontaneous polarization of the piezoelectric element may disappear, when an excessive voltage is applied to the piezoelectric element due to electrostatic discharge, for example. The piezoelectric element that has lost its spontaneous polarization is given spontaneous polarization again, when the piezoelectric element is supplied with a high voltage, for example.

[0005] The piezoelectric element may generate resonance of the gimbal due to a high voltage applied thereto. The resonance may cause the magnetic head to shake. For example, when the piezoelectric element generates resonance of the gimbal on a data zone of the magnetic disk, the magnetic head floating over the surface of the magnetic disk could be jolted, which possibly causes the magnetic head to come into contact with the data zone. If the magnetic head is prevented from coming into contact with the data zone, the reliability of the magnetic disk device can be improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is an exemplary diagram illustrating a configuration of a magnetic disk device according to a first embodiment;

[0007] FIG. 2 is an exemplary plan view illustrating a configuration of the magnetic disk device in the first embodiment;

[0008] FIG. 3 is an exemplary plan view illustrating a gimbal in the first embodiment;

[0009] FIG. 4 is an exemplary flowchart illustrating an example of an operation of the magnetic disk device in the first embodiment;

[0010] FIG. 5 is an exemplary flowchart illustrating an example of an operation of a magnetic disk device according to a second embodiment;

[0011] FIG. 6 is an exemplary flowchart illustrating an example of an operation of a magnetic disk device according to a third embodiment; and

[0012] FIG. 7 is an exemplary flowchart illustrating an example of an operation of a magnetic disk device according to a fourth embodiment.

DETAILED DESCRIPTION

[0013] In general, according to one embodiment, a magnetic disk device includes a magnetic disk, a gimbal, and a controller. The magnetic disk is configured to rotate around a first axis. The gimbal includes a magnetic head configured to perform recording and reproducing of data with respect to

the magnetic disk, and a piezoelectric element configured to be deformed in accordance with an applied drive voltage to move the magnetic head. The controller is configured to apply a polarization voltage to the piezoelectric element to give spontaneous polarization to the piezoelectric element. The polarization voltage satisfies a relationship of $\Delta V = (2\pi f)A \cdot \Delta T < \Delta V_{\max}$, where “ ΔV ” is a voltage change amount of the polarization voltage per unit time, “ f ” is a frequency of the polarization voltage, “ A ” is an amplitude of the polarization voltage, “ ΔT ” is a time resolution of the polarization voltage, and “ $\Delta V_{\max}$ ” is a voltage change amount per unit time that excites resonance of the gimbal in application to the piezoelectric element.

First Embodiment

[0014] Hereinafter, with reference to FIGS. **1** to **4**, a first embodiment will be explained. It should be noted that, in the present specification, there is a case where a plurality of expressions are used, with reference to a constituent element according to an embodiment and description on the element. The constituent elements and description thereon are mere examples, and are not limited to the expressions given in the present specification. A constituent element may be identified with a name different from that in the present specification. Further, a constituent element may be described by using an expression different from that in the present specification.

[0015] In the following explanation, “to prevent” is defined as, for example, to prevent the occurrence of an event, action, or influence, or to reduce the severity level of an event, action, or influence. Further in the following explanation, “to restrict” is defined as, for example, to prevent movement or rotation, or to allow movement or rotation within a specified range while preventing movement or rotation beyond that specified range.

[0016] FIG. **1** is an exemplary diagram illustrating a configuration of a magnetic disk device **10** according to the first embodiment. The magnetic disk device **10** is a hard disk drive (HDD), for example. However, the magnetic disk device **10** may be another magnetic disk device, such as a hybrid HDD.

[0017] The magnetic disk device **10** can be connected to a host **2**. The magnetic disk device **10** and the host **2** can communicate with each other in compliance with the Serial Attached SCSI (SAS) standard, for example. However, the standard of the communication path between the magnetic disk device **10** and the host **2** is not limited to this example.

[0018] The host **2** is a processor, personal computer, or server, for example. The magnetic disk device **10** can receive access commands (read commands and write commands) from the host **2**.

[0019] FIG. **2** is an exemplary plan view illustrating a configuration of the magnetic disk device **10** in the first embodiment. As illustrated in FIGS. **1** and **2**, the magnetic disk device **10** includes a spindle motor (SPM) **11**, a plurality of magnetic disks **12**, a plurality of magnetic heads **13**, an actuator **14**, a voice coil motor (VCM) **15**, a plurality of micro-actuators (MA) **16**, a ramp load mechanism **17**, an outer stopper **21**, an inner stopper **22**, a system-on-chip (SoC) **23**, a head amplifier **24**, and a servo controller (SVC) **25**.

[0020] The magnetic heads **13** may also be referred to as sliders. Each of the MAs **16** is an example of a piezoelectric element. The ramp load mechanism **17** is an example of a ramp. The inner stopper **22** is an example of a stopper. The SVC **25** is an example of a controller. The SVC **25** may also be referred to as a servo combo.

[0021] As illustrated in FIG. **1**, the magnetic disks **12** are held on the hub of the SPM **11** by a clamp, for example. The SPM **11** can rotate the magnetic disks **12** together around a central axis A_{xd} . In other words, the magnetic disks **12** can rotate around the central axis A_{xd} . The central axis A_{xd} is an example of a first axis, and is the central axis of the magnetic disks **12**, for example.

[0022] As illustrated in FIG. **2**, each of the plurality of magnetic disks **12** includes a recording surface **12s**. For example, each magnetic disk **12** is provided with recording surfaces **12s** one on either side thereof. The recording surface **12s** includes a data zone **12sd**, an inner peripheral zone **12si**, and an outer peripheral zone **12so**.

[0023] The data zone **12sd** is positioned between the inner peripheral zone **12si** and the outer peripheral zone **12so**, and data is recorded therein. The inner peripheral zone **12si** is closer to the central axis **Axd** than the data zone **12sd** and the outer peripheral zone **12so**. The outer peripheral zone **12so** is more distant from the central axis **Axd** than the data zone **12sd** and the inner peripheral zone **12si**.

[0024] The number of the magnetic heads **13** is set such that the plurality of magnetic heads **13** can access the recording surfaces **12s** of the magnetic disks **12**. Each of the magnetic heads **13** can be placed on corresponding one of the recording surfaces **12s**. The magnetic head **13** floats over the recording surface **12s** while the magnetic disk **12** rotates. Each of the plurality of magnetic heads **13** can perform data recording and data reproducing with respect to the data zone **12sd** of the corresponding one of the recording surfaces **12s**.

[0025] The actuator **14** includes a carriage **31** and a plurality of suspensions **32**. The actuator **14** may include a plurality of carriages **31**. The carriage **31** includes a block **41**, a plurality of arms **42**, and a holder **43**.

[0026] The block **41** can rotate around a central axis **Axc**. The central axis **Axc** extends substantially in parallel to the central axis **Axd** and is distant from the central axis **Axd**. The number of arms **42** is set in accordance with the number of magnetic heads **13**. The arms **42** protrude from the block **41** in a direction substantially perpendicular to the central axis **Axc**. The arms **42** extend substantially in parallel to each other. The holder **43** protrudes from the block **41** and is disposed on the side opposite to the arms **42**.

[0027] The number of suspensions **32** is set in accordance with the number of magnetic heads **13**. Each of the suspensions **32** includes a base plate **45**, a load beam **46**, and a flexure **47**. The base plate **45** and the load beam **46** are made of stainless steel, for example.

[0028] The base plate **45** is attached to the tip of the arm **42**. The load beam **46** is formed in the shape of a thin plate that can be elastically deformed. The load beam **46** is attached to the base plate **45** so as to extend from the base plate **45**. The load beam **46** includes a lift tab **46a** provided at the tip of the actuator **14**.

[0029] The flexure **47** is a kind of flexible printed circuit board (FPC) and includes a plurality of insulating layers, a conductive layer, and a backing plate. The conductive layer is provided between two insulating layers, for example, and includes a plurality of wiring lines and a plurality of pads. The backing plate is made of stainless steel, for example, and reinforces the flexure **47**.

[0030] The backing plate of the flexure **47** is attached to the load beam **46** by welding, for example. Each of the magnetic heads **13** is mounted on the flexure **47** near the tip of the corresponding one of the suspensions **32**.

[0031] The VCM **15** includes a voice coil, a pair of yokes, and a magnet attached to the yokes. The voice coil is held on the holder **43**. The VCM **15** can rotate the actuator **14** around the central axis **Axc** within a specified range. The actuator **14** rotates around the central axis **Axc** to move the magnetic heads **13** on the suspensions **32** with respect to the magnetic disks **12**.

[0032] The VCM **15** can rotate actuator **14** in a first direction **Dr1** and a second direction **Dr2** around the central axis **Axc**. The second direction **Dr2** is a direction opposite to the first direction **Dr1**. When the actuator **14** rotates in the first direction **Dr1**, the magnetic heads **13** are moved away from the central axis **Axd**. When the actuator **14** rotates in the second direction **Dr2**, the magnetic heads **13** are moved closer to the central axis **Axd**.

[0033] The VCM **15** moves the suspensions **32** and the arms **42** substantially in parallel to the recording surfaces **12s** of the magnetic disks **12**. In this embodiment, the VCM **15** moves the magnetic heads **13**, with respect to the magnetic disks **12**, substantially in a radial direction perpendicular to the **Axd**.

[0034] FIG. 3 is an exemplary plan view illustrating a gimbal **G** in the first embodiment. As illustrated in FIG. 3, the magnetic disk device **10** is provided with gimbals **G** which each adjust the position of corresponding one of the magnetic heads **13**. Each of the gimbals **G** includes one of the

magnetic heads **13**, two or more of the MAs **16**, and a part of one of the suspensions **32**.

[0035] The MAs **16** are mounted on the suspension **32**. For example, two MAs **16** are mounted on one suspension **32**. In this embodiment, the MAs **16** are mounted on the flexure **47** near the magnetic head **13**. However, the MAs **16** are not limited to this example. For example, the MAs **16** may be disposed near the connection between the base plate **45** and the load beam **46**.

[0036] Each of the MAs **16** includes a piezoelectric body **16a** and two electrodes **16b** and **16c**. The electrodes **16b** and **16c** are provided on the surface of the piezoelectric body **16a** and are distant from each other. The electrode **16b** is electrically connected to the SVC **25** through the flexure **47**. The electrode **16c** is electrically connected to the load beam **46**, for example, through a conductive layer and the backing plate of the flexure **47**, and is set to the ground potential. While a voltage is applied to the electrode **16b**, the MA **16** is deformed due to the inverse piezoelectric effect of the piezoelectric body **16a**. For example, the piezoelectric body **16a** expands and contracts in the longitudinal direction of the suspension **32**.

[0037] The two MAs **16** on the suspension **32** expand and contract individually to bend the suspension **32** substantially in parallel to the recording surface **12s**. The magnetic head **13** on the flexure **47** of the suspension **32** is moved with respect to the magnetic disk **12** by the deformation of the MAs **16**.

[0038] As illustrated in FIG. 2, the lift tabs **46a** of the actuator **14** can be moved along a trajectory T with respect to the recording surfaces **12s**. The ramp load mechanism **17** is provided on the trajectory T near the outer end of the magnetic disks **12**. The central axis Axd of the magnetic disks **12** is positioned near the trajectory T. The actuator **14** allows the lift tabs **46a** to move between the ramp load mechanism **17** and the central axis Axd.

[0039] The ramp load mechanism **17** is distant from the central axis Axd in the first direction Dr1. The ramp load mechanism **17** is used to park the magnetic heads **13** in the unloading time, for example. For example, the ramp load mechanism **17** can support the lift tabs **46a** of the load beams **46** to hold the actuator **14** and thereby hold the magnetic heads **13** in a retreat position.

[0040] The outer stopper **21** and the inner stopper **22** are distant from each other around the central axis Axc. The actuator **14** is positioned between the outer stopper **21** and the inner stopper **22**.

[0041] When the actuator **14** rotates to a specified position in the first direction Dr1, the actuator **14** is brought into contact with the outer stopper **21**. The outer stopper **21** is brought into contact with the actuator **14** to restrict the actuator **14** from rotating in the first direction Dr1. While the actuator **14** is in contact with the outer stopper **21**, the ramp load mechanism **17** holds the actuator **14**, and the magnetic heads **13** are positioned in the retreat position.

[0042] When the actuator **14** rotates to a specified position in the second direction Dr2, the actuator **14** is brought into contact with the inner stopper **22**. The inner stopper **22** is brought into contact with the actuator **14** to restrict the actuator **14** from rotating in the second direction Dr2. While the actuator **14** is in contact with the inner stopper **22**, the magnetic heads **13** are positioned on the inner peripheral zones **12si**. In other words, each magnetic head **13** is positioned outside the data zone **12sd**.

[0043] As illustrated in FIG. 1, the SoC **23** includes a Micro-Processing Unit (MPU) **51** and a hard disk controller (HDC) **52**. The MPU **51** may also be referred to as a processor.

[0044] The MPU **51** operates in accordance with a firmware program. The firmware program is stored in a specified non-volatile storage area. The storage area may be provided by a part of the magnetic disks **12** or may be provided in a read-only memory (ROM) of the SoC **23**. The MPU **51** controls the overall operation of the magnetic disk device **10**. For example, the MPU **51** controls the HDC **52** and the SVC **25**.

[0045] The HDC **52** controls sending and receiving of data performed to and from the host **2**. The HDC **52** interprets access commands from the host **2** and, on the basis of the interpretation results, executes various kinds of control, such as access to the magnetic disks **12**. Specifically, the HDC **52** controls the access to the magnetic disks **12** using the magnetic heads **13** via the read/write

channels (RWC) of the SoC **23** and the head amplifier **24**.

[0046] The head amplifier **24** amplifies signals read by the magnetic heads **13** from the magnetic disks **12**, and outputs and supplies the signals to the RWC of the SoC **23**. The RWC demodulates the signals supplied from the head amplifier **24** to digital data and supplies this data to the HDC **52**.

[0047] Further, the head amplifier **24** is supplied with signals corresponding to the digital data, from the RWC. The head amplifier **24** amplifies the signals supplied from the RWC and supplies the signals to the magnetic heads **13**. Each of the magnetic heads **13** records the signals supplied from the head amplifier **24**, into the data zone **12sd** of the corresponding magnetic disk **12**.

[0048] The SVC **25** performs drive control over the SPM **11**, the VCM **15**, and the MAs **16**.

Specifically, the SVC **25** drives the VCM **15** and the MAs **16** on the basis of instructions from the MPU **51**, and thereby sets each of the magnetic heads **13** at the position instructed by the MPU **51**.

[0049] The MPU **51** calculates an instruction value of the drive voltage for the VCM **15** and an instruction value of the drive voltage for each MA **16**, to cause the position of each magnetic head **13** to follow the target position. The MPU **51** calculates each of the instruction values while using, as a feedback input, the position signal read by the magnetic head **13** from the servo information formed on the recording surface **12s** of the magnetic disk **12**, and transmits each of the instruction values thereby obtained to the SVC **25**.

[0050] The SVC **25** supplies the VCM **15** with a voltage according to the instruction value of the drive voltage for the VCM **15**, and supplies each MA **16** with a voltage according to the instruction value of the drive voltage for the MA **16**. Consequently, the magnetic head **13** is set at the target position.

[0051] The SVC **25** supplies the electrode **16b** of each MA **16** with a drive voltage corresponding to the instruction value. The piezoelectric body **16a** of the MA **16** is deformed by the potential difference between the electrode **16b** supplied with the drive voltage and the electrode **16c** set to the ground potential. In other words, the MA **16** is deformed in accordance with the drive voltage applied to the electrode **16b**. Consequently, the position of the magnetic head **13** is adjusted.

[0052] The SVC **25** executes the control over loading/unloading of the actuator **14** by driving the VCM **15** on the basis of instructions from the MPU **51**. For the unloading, the SVC **25** drives the VCM **15** to rotate the actuator **14** in the first direction Dr1. Consequently, the lift tabs **46a** of the load beams **46** are supported by the ramp load mechanism **17**, and the magnetic heads **13** are moved away from the magnetic disks **12**. For the loading, the SVC **25** drives the VCM **15** to rotate the actuator **14** in the second direction Dr2. Consequently, the lift tabs **46a** of the load beam **46** are moved away from the ramp load mechanism **17**, and the magnetic heads **13** are placed on the recording surfaces **12s** of the magnetic disks **12**.

[0053] The SVC **25** drives the SPM **11** on the basis of instructions from the MPU **51**. The SVC **25** drives the SPM **11** such that the rotation speed of the SPM **11** is substantially constant at a predetermined target speed.

[0054] In the unloading, the ramp load mechanism **17** is brought into contact with the lift tabs **46a** of the load beams **46**. The surface of the ramp load mechanism **17** in contact with the lift tabs **46a** is made of, for example, a synthetic resin. For this reason, due to friction between the surface of the ramp load mechanism **17** and the lift tabs **46a**, the ramp load mechanism **17** may be charged with static electricity.

[0055] When the ramp load mechanism **17** generates electrostatic discharge (ESD), static electricity flows from the lift tabs **46a** to the load beams **46**. Each of the load beams **46** is connected to the backing plate of the flexure **47** by welding. Further, the backing plate of the flexure **47** is electrically connected to the electrode **16c** of each MA **16** through a via and a conductive layer of the flexure **47**. Therefore, the static electricity is applied to the electrode **16c**.

[0056] When an excessive voltage is applied to an MA **16** due to the ESD, the spontaneous polarization of the MA **16** may disappear or alter. For example, the orientations of the spontaneous polarization of a plurality of crystals of the piezoelectric body **16a** may become random or may

become different from the desired orientations. In this embodiment, the SVC 25 applies a voltage to each MA 16, and thereby gives spontaneous polarization in the desired orientations to the MA 16 again.

[0057] As illustrated in FIG. 3, the magnetic disk device 10 further includes a resistor 55. For example, the resistor 55 is provided in a circuit that drives the MAS 16, and is disposed in wiring that connects the SVC 25 to the electrode 16b of each MA 16. For example, the SoC 23 checks the current between the resistor 55 and the SVC 25, and the current between the resistor 55 and the electrode 16b, and detects the occurrence of ESD on the basis of these currents. It is noted that, the magnetic disk device 10 may detect the occurrence of ESD by another method.

[0058] FIG. 4 is an exemplary flowchart illustrating an example of an operation of the magnetic disk device 10 in the first embodiment. With reference to FIG. 4, an explanation will be given below of an example of an operation of repolarization of each MA 16 in this embodiment.

[0059] First, the MPU 51 drives the VCM 15 via the SVC 25 to perform unloading control (S101). As the VCM 15 rotates the actuator 14 in the first direction Dr1, the lift tabs 46a of the load beams 46 are supported by the ramp load mechanism 17. This unloading control is a type of control performed during the normal operation of the magnetic disk device 10, regardless of the occurrence of ESD. It is noted that, the unloading control is not limited to this example.

[0060] Then, the MPU 51 drives the VCM 15 via the SVC 25 to perform loading control (S102). As the VCM 15 rotates the actuator 14 in the second direction Dr2, the actuator 14 which has been held by the ramp load mechanism 17 is moved from the ramp load mechanism 17 in the second direction Dr2, and the magnetic head 13 lands on the recording surfaces 12s of the magnetic disks 12.

[0061] Then, the MPU 51 drives the VCM 15 via the SVC 25, and performs seeking (movement) until the actuator 14 is brought into contact with the inner stopper 22 (S103). During the period from S101 to S103, the SVC 25 does not drive the MAs 16. It is noted that, the operation of the SVC 25 is not limited to this example.

[0062] For example, in the unloading control and loading control described above, ESD may occur due to friction between the surface of the ramp load mechanism 17 and the lift tabs 46a. The SoC 23 determines whether ESD has been detected (S104). The SoC 23 checks the occurrence of ESD periodically or during the period from S101 to S103.

[0063] When the occurrence of ESD has been detected by the time of S104 (S104: Yes), the SoC 23 transmits a repolarization command to the SVC 25. It is noted that, the SoC 23 may transmit the repolarization command to the SVC 25 before S101 or during the period from S101 to S103.

[0064] When the SVC 25 receives the repolarization command from the SoC 23, the SVC 25 applies a polarization voltage to the electrode 16b of each MA 16 while the actuator 14 is in contact with the inner stopper 22 (S105). In other words, when static electricity is applied to the MAs 16, the SVC 25 brings the actuator 14 into contact with the inner stopper 22, and applies the polarization voltage to the electrodes 16b.

[0065] The polarization voltage is a DC bias voltage or a voltage formed of a DC bias voltage superimposed with an AC voltage. The polarization voltage is higher than the drive voltage. Further, the polarization voltage is equal to or greater than the absolute value of a voltage that causes the spontaneous polarization of each MA 16 to disappear.

[0066] The SVC 25 applies the polarization voltage to the electrode 16b of each MA 16 to give spontaneous polarization to the MAs 16. Specifically, when the polarization voltage is applied to the electrode 16b, the orientations of the spontaneous polarization of a plurality of crystals in the piezoelectric body 16a become substantially the same. Consequently, the MA 16 obtains spontaneous polarization again. When a drive voltage is applied to the electrode 16b, the MA 16 performs desired deformation due to the inverse piezoelectric effect.

[0067] The polarization voltage has a specified voltage, amplitude, and frequency. Further, the SVC 25 applies the polarization voltage to the electrode 16b for a specified period of time. The

voltage, amplitude, frequency, and time are set to values capable of giving spontaneous polarization to the MA **16**.

[0068] The voltage, amplitude, frequency, and time of the polarization voltage are set to satisfy the following formula (1) and formula (2), for example. It is noted that, the polarization voltage is not limited to this example.

$$[00001] \quad V = (2 \ f)A \cdot \text{Math.} \quad T < \ V_{\max} \quad (1) \quad T = N / f < T_{\max} \quad (2)$$

[0069] In the formula (1) and formula (2) described above, “ ΔV ” is the voltage change amount of the polarization voltage per unit time, “ f ” is the frequency of the polarization voltage, “ A ” is the amplitude of the polarization voltage, “ ΔT ” is the time resolution of the polarization voltage, “ T ” is the time in which the polarization voltage is applied to the electrode **16b**, “ N ” is the number of cycles with which the polarization voltage is applied to the electrode **16b**, and “ $T_{\max}$ ” is the start-up time of the positioning control of the magnetic head **13** by the MAs **16**. The time resolution of the polarization voltage corresponds to the control sampling period of the MAs **16**.

[0070] In the formula (1) described above, “ $\Delta V_{\max}$ ” is the voltage change amount per unit time that excites the main resonance of the gimbal G in application to the electrode **16b** of the MA **16**. This $\Delta V_{\max}$ is set on the basis of experiments, for example.

[0071] When the main resonance of the gimbal G is generated, the magnetic head **13** which floats over the magnetic disk **12** may be caused to shake (in the form of pitching) that repeats approaching and receding with respect to the magnetic disk **12**. For example, when a voltage whose voltage change amount per unit time is $\Delta V_{\max}$ is applied to the electrode **16b**, the gimbal G causes the magnetic head **13** to generate pitching that may bring the magnetic head **13** into contact with the magnetic disk **12**. In this embodiment, the SVC **25** supplies the electrode **16b** with a polarization voltage that satisfies the formula (1), so as to prevent the main resonance of the gimbal G and thereby to prevent the magnetic head **13** from being brought into contact with the magnetic disk **12**.

[0072] The magnetic head **13** may be allowed to generate pitching. When the actuator **14** is in contact with the inner stopper **22**, the magnetic head **13** is positioned on the inner peripheral zone **12si** of the magnetic disk **12**. Even if the magnetic head **13** is brought into contact with the inner peripheral zone **12si**, no contact is made with the data zone **12sd**. Thus, the magnetic head **13** is prevented from damaging the data recorded in the data zone **12sd**.

[0073] After the polarization voltage is applied to each MA **16**, or when no ESD has been detected in **S104** (**S104**: No), the MPU **51** controls the SVC **25** to start positioning control of each magnetic head **13** by the MAs **16** (**S106**).

[0074] The SVC **25** applies the polarization voltage to the electrode **16b** for a period of time that satisfies the formula (2). Accordingly, as soon as the positioning control of each magnetic head **13** by the MAs **16** becomes able to start, the MPU **51** can start this positioning control. Further, as the time for applying the polarization voltage to the electrode **16b** is set shorter, the performance of the magnetic disk device **10** can be improved.

[0075] Then, the SVC **25** drives the VCM **15** to rotate the actuator **14** in the first direction **Dr1**. Consequently, the actuator **14** is moved away from the inner stopper **22**, and each magnetic head **13** is moved from the inner peripheral zone **12si** to the data zone **12sd** (on-track) (**S107**). As a result, the magnetic disk device **10** can adjust the position of each magnetic head **13** by the MAs **16**, which have been given the desired spontaneous polarization again.

[0076] In the magnetic disk device **10** according to the first embodiment described above, the magnetic disks **12** is configured to rotate around the central axis **Axd**. The gimbal G includes the magnetic head **13** and the MA **16**. The magnetic head **13** is configured to perform recording and reproducing of data with respect to the magnetic disk **12**. The MA **16** is configured to be deformed in accordance with an applied drive voltage to move the magnetic head **13**. The SVC **25** is configured to apply a polarization voltage to the MA **16** to give spontaneous polarization to the MA

16. The polarization voltage satisfies a relationship of $\Delta V = (2\pi f)A \cdot \Delta T < \Delta V_{\max}$, where “ ΔV ” is the voltage change amount of the polarization voltage per unit time, “ f ” is the frequency of the polarization voltage, “ A ” is the amplitude of the polarization voltage, “ ΔT ” is the time resolution of the polarization voltage, and “ $\Delta V_{\max}$ ” is the voltage change amount per unit time that excites the resonance of the gimbal **G** in application to the MA **16**.

[0077] When an excessive voltage is applied to an MA **16** due to ESD, for example, the spontaneous polarization of the MA **16** may disappear. The SVC **25** can give spontaneous polarization to the MA **16** again by applying the polarization voltage to the electrode **16b** of the MA **16**. Therefore, the magnetic disk device **10** can recover the spontaneous polarization of the MA **16** even if the spontaneous polarization of the MA **16** temporarily disappears, and can thereby adjust the position of each magnetic head **13** more accurately by the MAs **16**. In addition, the voltage change amount ΔV per unit time of the polarization voltage is lower than the voltage change amount $\Delta V_{\max}$ per unit time that excites the resonance of the gimbal **G**. Consequently, the magnetic disk device **10** can prevent the magnetic head **13** from being caused to shake that may bring the magnetic head **13** contact with the magnetic disk **12**, while the polarization voltage is being applied to the electrode **16b** of the MA **16**. As a result, it is possible to improve the reliability of the magnetic disk device **10**.

[0078] The polarization voltage satisfies a relationship of $T = N/f < T_{\max}$, where “ T ” is the time in which the polarization voltage is applied to the MA **16**, “ N ” is the number of cycles with which the polarization voltage is applied to the MA **16**, and “ $T_{\max}$ ” is the start-up time.

[0079] The SVC **25** applies the polarization voltage to the electrode **16b** of the MA **16** for a time shorter than the start-up time. Thus, the magnetic disk device **10** can give spontaneous polarization to the MA **16** before the magnetic head **13** starts recording and reproducing of data with respect to the magnetic disk **12**. Therefore, the magnetic disk device **10** can prevent the time from being extended until the magnetic head **13** starts recording and reproducing of data with respect to the magnetic disk **12**, and can adjust the position of the magnetic head **13** by the MA **16** more reliably.

[0080] The actuator **14** is configured to rotate around the central axis A_{xc} to move the magnetic heads **13** with respect to the magnetic disks **12**. The ramp load mechanism **17** is distant from the central axis A_{xd} in the first direction $Dr1$ around the central axis A_{xc} , and configured to hold the actuator **14**. The inner stopper **22** is configured to be brought into contact with the actuator **14** to restrict the actuator **14** from rotating in the second direction $Dr2$ opposite to the first direction $Dr1$. The SVC **25** is configured to apply the polarization voltage to the electrode **16b** when the actuator **14** is in contact with the inner stopper **22**.

[0081] When the actuator **14** is in contact with the inner stopper **22**, each magnetic head **13** is positioned outside the data zone **12sd** of the magnetic disk **12**. Therefore, even if the MA **16** supplied with the polarization voltage to the electrode **16b** excites the resonance of the gimbal **G**, the magnetic disk device **10** can prevent the magnetic head **13** from being brought into contact with the data zone **12sd** of the magnetic disk **12**, thereby improving the reliability.

[0082] The SVC **25** is configured to bring the actuator **14** into contact with the inner stopper **22**, and apply the polarization voltage to the MA **16**, when static electricity is applied to the MA **16**.

[0083] For example, when the actuator **14** which has been held by the ramp load mechanism **17** is moved, the ramp load mechanism **17** may be charged with static electricity due to friction between the ramp load mechanism **17** and the actuator **14**. When ESD is generated, the static electricity of the ramp load mechanism **17** is applied to the electrode **16c** of the MA **16** through the load beam **46** of the actuator **14** and the backing plate and wiring line of the flexure **47**, for example. This static electricity may cause the spontaneous polarization of the MA **16** to disappear. However, in this embodiment, when the static electricity is applied to the MA **16**, the SVC **25** applies the polarization voltage to the electrode **16b** of the MA **16**. Consequently, the magnetic disk device **10** can recover the spontaneous polarization of the MA **16** efficiently.

Second Embodiment

[0084] Hereinafter, with reference to FIG. 5, a second embodiment will be explained. It should be noted that, in the following description of a plurality of embodiments, the constituent elements having functions substantially the same as those of constituent elements previously described are denoted by reference symbols the same as those for the constituent elements previously described, and, further, their description may be omitted in some cases. Furthermore, a plurality of constituent elements denoted by the same reference symbols are not necessarily the same in all the functions and properties, but may be different in function and/or property as needed for the respective embodiments.

[0085] FIG. 5 is an exemplary flowchart illustrating an example of an operation of a magnetic disk device **10** according to the second embodiment. With reference to FIG. 5, an explanation will be given below of an example of an operation of repolarization of each MA **16** in the second embodiment. In the second embodiment, the magnetic disk device **10** gives spontaneous polarization to each MA **16** again when ESD is detected.

[0086] First, the SoC **23** determines whether ESD has been detected (**S201**). When no ESD has been detected (**S201**: No), the SoC **23** repeats **S201** until ESD is detected.

[0087] When the SoC **23** has detected ESD (**S201**: Yes), the HDC **52** transmits a repolarization command to the SVC **25**. In response to the repolarization command, the SVC **25** drives the VCM **15** and performs the seeking until the actuator **14** is brought into contact with the inner stopper **22** (**S103**).

[0088] Then, the SVC **25** applies the polarization voltage to the electrode **16b** of each MA **16** while the actuator **14** is in contact with the inner stopper **22** (**S105**). In other words, when static electricity is applied to an MA **16**, the SVC **25** brings the actuator **14** in contact with the inner stopper **22**, and applies the polarization voltage to the electrode **16b**. Thereafter, **S106** and **S107** are executed, as in the first embodiment.

[0089] In the magnetic disk device **10** of the second embodiment described above, the SVC **25** is configured to bring the actuator **14** into contact with the inner stopper **22**, and apply the polarization voltage to the MA **16**, when static electricity is applied to the MA **16**. Consequently, the magnetic disk device **10** can recover the spontaneous polarization of the MA **16** efficiently.

Third Embodiment

[0090] Hereinafter, with reference to FIG. 6, a third embodiment will be explained. FIG. 6 is an exemplary flowchart illustrating an example of an operation of a magnetic disk device **10** according to the third embodiment. With reference to FIG. 6, an explanation will be given below of an example of an operation of repolarization of each MA **16** in the third embodiment. In the third embodiment, the magnetic disk device **10** gives spontaneous polarization to each MA **16** again in the loading/unloading time.

[0091] As in the first embodiment, the MPU **51** performs the unloading control of **S101**, the loading control of **S102**, and the seeking control down to the inner stopper **22** of **S103**. When the actuator **14** is brought into contact with the inner stopper **22**, the HDC **52** transmits a repolarization command to the SVC **25**.

[0092] When the SVC **25** receives the repolarization command from the HDC **52**, the SVC **25** applies the polarization voltage to the electrode **16b** of each MA **16** while the actuator **14** is in contact with the inner stopper **22** (**S105**). Specifically, when the actuator **14** which has been held by the ramp load mechanism **17** is moved from the ramp load mechanism **17** in the second direction **Dr2**, the SVC **25** brings the actuator **14** into contact with the inner stopper **22**, and applies the polarization voltage to the electrode **16b**. Thereafter, **S106** and **S107** are executed, as in the first embodiment.

[0093] In the magnetic disk device **10** of the third embodiment described above, the SVC **25** is configured to bring the actuator **14** into contact with the inner stopper **22**, and apply the polarization voltage to the MA **16**, when the actuator **14** is moved from the ramp load mechanism **17** in the second direction **Dr2**.

[0094] For example, in a case where the polarization voltage is applied to the electrode **16b** of the MA **16** while the actuator **14** is being held by the ramp load mechanism **17**, there is a possibility that the spontaneous polarization of the MA **16** disappears due to ESD before the actuator **14** is separated from the ramp load mechanism **17**. However, the SVC **25** applies the polarization voltage to the electrode after the actuator **14** has separated from the ramp load mechanism **17** and been brought into contact with the inner stopper **22**. Consequently, the magnetic disk device **10** can prevent the spontaneous polarization of each MA **16** from disappearing again.

Fourth Embodiment

[0095] Hereinafter, with reference to FIG. 7, a fourth embodiment will be explained. FIG. 7 is an exemplary flowchart illustrating an example of an operation of a magnetic disk device **10** according to the fourth embodiment. With reference to FIG. 7, an explanation will be given below of an example of an operation of repolarization of each MA **16** in the fourth embodiment. In the fourth embodiment, the magnetic disk device **10** gives spontaneous polarization to each MA **16** again at constant time intervals.

[0096] First, the HDC **52** determines whether a specified time has elapsed (**S401**). When the specified time has not elapsed (**S401**: No), the HDC **52** repeats **S401** until the specified time has elapsed.

[0097] When the specified time has elapsed (**S401**: Yes), the HDC **52** transmits a repolarization command to the SVC **25**. In response to the repolarization command, the SVC **25** drives the VCM **15** and performs the seeking (movement) until the actuator **14** is brought into contact with the inner stopper **22** (**S103**). It is noted that, when the magnetic head **13** is accessing the magnetic disk **12**, the MPU **51** may perform the seeking control after this access is completed.

[0098] Then, the SVC **25** applies the polarization voltage to the electrode **16b** of each MA **16** while the actuator **14** is in contact with the inner stopper **22** (**S105**). In other words, at constant time intervals, the SVC **25** brings the actuator **14** into contact with the inner stopper **22** and applies the polarization voltage to the electrode **16b**. Thereafter, **S106** and **S107** are executed, as in the first embodiment.

[0099] In the magnetic disk device **10** of the fourth embodiment described above, at constant time intervals, the SVC **25** brings the actuator **14** into contact with the inner stopper **22**, and applies the polarization voltage to the MA **16**. Consequently, the magnetic disk device **10** can periodically recover the spontaneous polarization of the MA **16**, thereby improving the reliability.

[0100] In each of the embodiments described above, the polarization voltage is used to give spontaneous polarization again to the MAs **16** that have lost the desired spontaneous polarization due to ESD. However, the polarization voltage is not limited to this example. For example, the polarization voltage may be used to give spontaneous polarization to the MAs **16** that have lost the desired spontaneous polarization due to another reason.

[0101] Further, in each of the plurality of embodiments described above, the SVC **25** brings the actuator **14** into contact with the inner stopper **22** and applies the polarization voltage to each electrode **16b**. However, the SVC **25** may apply the polarization voltage to each electrode **16b**, when the actuator **14** is separated from the inner stopper **22** and each magnetic head **13** is positioned on the data zone **12sd** or the outer peripheral zone **12so**, or when the magnetic heads **13** are positioned away from the magnetic disks **12**.

[0102] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A magnetic disk device comprising: a magnetic disk configured to rotate around a first axis; a gimbal that includes a magnetic head configured to perform recording and reproducing of data with respect to the magnetic disk, and a piezoelectric element configured to be deformed in accordance with an applied drive voltage to move the magnetic head; and a controller configured to apply a polarization voltage to the piezoelectric element to give spontaneous polarization to the piezoelectric element, wherein the polarization voltage satisfies a relationship of $\Delta V = (2\pi f)A \cdot \Delta T < \Delta V_{\max}$, where “ ΔV ” is a voltage change amount of the polarization voltage per unit time, “ f ” is a frequency of the polarization voltage, “ A ” is an amplitude of the polarization voltage, “ ΔT ” is a time resolution of the polarization voltage, and “ $\Delta V_{\max}$ ” is a voltage change amount per unit time that excites resonance of the gimbal in application to the piezoelectric element.
2. The magnetic disk device according to claim 1, wherein the polarization voltage satisfies a relationship of $T = N/f < T_{\max}$, where “ T ” is a time in which the polarization voltage is applied to the piezoelectric element, “ N ” is a number of cycles with which the polarization voltage is applied to the piezoelectric element, and “ $T_{\max}$ ” is a start-up time.
3. The magnetic disk device according to claim 1, further comprising: an actuator configured to rotate around a second axis to move the magnetic head with respect to the magnetic disk; a ramp distant from the first axis in a first direction around the second axis, and configured to hold the actuator; and a stopper configured to be brought into contact with the actuator to restrict the actuator from rotating in a second direction opposite to the first direction; wherein the controller is configured to apply the polarization voltage to the piezoelectric element when the actuator is in contact with the stopper.
4. The magnetic disk device according to claim 3, wherein the controller is configured to bring the actuator into contact with the stopper and apply the polarization voltage to the piezoelectric element, when static electricity is applied to the piezoelectric element.
5. The magnetic disk device according to claim 3, wherein the controller is configured to bring the actuator into contact with the stopper and apply the polarization voltage to the piezoelectric element, at constant time intervals.
6. The magnetic disk device according to claim 3, wherein the controller is configured to bring the actuator into contact with the stopper and apply the polarization voltage to the piezoelectric element, when the actuator is moved from the ramp in the second direction.
7. The magnetic disk device according to claim 1, wherein the polarization voltage has a DC bias voltage, or a voltage formed of a DC bias voltage superimposed with an AC voltage.
8. The magnetic disk device according to claim 1, wherein the polarization voltage is higher than the drive voltage.
9. The magnetic disk device according to claim 1, wherein the polarization voltage is equal to or greater than an absolute value of a voltage that causes spontaneous polarization of the piezoelectric element to disappear.
10. A magnetic disk device comprising: a magnetic disk configured to rotate around a first axis; a gimbal that includes a magnetic head configured to perform recording and reproducing of data with respect to the magnetic disk, and a piezoelectric element configured to be deformed in accordance with an applied drive voltage to move the magnetic head; an actuator configured to rotate around a second axis to move the magnetic head with respect to the magnetic disk; a ramp distant from the first axis in a first direction around the second axis, and configured to hold the actuator; a stopper configured to be brought into contact with the actuator to restrict the actuator from rotating in a second direction opposite to the first direction; and a controller configured to apply a polarization

voltage to the piezoelectric element to give spontaneous polarization to the piezoelectric element, when the actuator is brought into contact with the stopper.
